

Capstone Project Final Report
ECOM30002/90002 Econometrics 2

Gender's Effect on Individuals' Education:
How Does Gender Affect Educational Gaps in Tanzania?

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Abstract

In this report I examine the causal influence of gender on the amount of schooling completed by boys and girls in Tanzania. While I establish an aggregate causal effect of gender, I struggle to quantify the effects of specific channels through which gender affects educational gaps due to the cross-sectional nature of the data. This report provides a framework for which to examine mediator effects and analyses statistical relationships within subsamples.

Introduction

Studies have consistently documented that an additional year of education in sub-Saharan Africa returns on average a 10.5% increase in private and social returns (Psacharopoulos & Patrinos, 2018). Educational gaps provide information about the magnitude of education deficits and identify groups of individuals to whom different policy options should be proposed. The purpose of this study is to assess gender's impact on educational gaps experienced in Tanzania and explore several channels of gender's effect. Sex inequalities are evident: compared to 15% of boys, approximately 19% of girls of primary school age are out of school. This gap widens to 51% and 61% respectively at secondary school (World Bank, 2018).

The study utilises a simple ordinary least squares (OLS) model to identify the overall relationship that exists between gender and educational gaps. The research investigates the degree of heterogeneity in OLS models as well as mediator variables using a two-stage least squares (2SLS) technique, utilising household food insecurity and household wall material as instrumental variables.

Our results suggest that there are gender effects on an individual's educational gap that vary within age groups and I am also able to measure agricultural participation as a channel through which household heads are affected by an educational gender bias. Such causal and decompositional interpretations face limitations however, due to the cross sectional nature of the variables, when educational outcomes are a reflection of a lifetime of educational decisions.

Data

Data Source

- Tanzania National Panel Survey (NPS) Wave 5 (2020–21)
- Sample size: 11685 individuals
- Missing value treatment: observations with non-missing values across all variables

Variable Construction

- Dependent variable: $years_{gap_i} = benchmark - completed\ years\ schooling_i$

where benchmark equals age minus 7 and is capped at 18. Therefore, higher values indicate larger education shortfalls.

- Explanatory variable: $female_i$

is a dummy variable equal to 1 for females and 0 for males.

Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
$years_{gap_i}$	10.1949	4.6980	-1	18
$female_i$	0.5138	0.4998	0	1
age_i	28.0227	17.7082	7	95

The mean value of $years_{gap}$ is 10.1949 years (SD = 4.6980), showing wide variation in educational shortfalls across the sample. The mean of $females$ is 0.5138, a near-equal gender split. The wide range (7 to 95) starts at the compulsory age for education at 7 years old.

Control Variable: Age is simply expressed in years. Age may be correlated with the error term due to cohort effects and have sampling bias due to survivorship bias.

Variable	Simple OLS Regression	Multiple OLS Regression
female	0.9890*** (0.08658)	0.7747*** (0.071301)
Constant	9.6729*** (0.06290)	4.0010*** (0.091938)
age		0.1592*** (0.002132)
N	11685	11685
R ²	0.01096	0.3303

Methodology - Causal OLS

Gender is treated as a predetermined characteristic, consistent with the assumption of random assignment of biological sex at birth evidenced in Angrist & Evans (1998), and thus I am able to interpret the OLS model causally. I aim to find how gender affects education through specific mediating channels, not the effect of being biologically male or female. The endogeneity of these unspecified variables function as mediator variables rather than sources of omitted variable bias.

Simple and Multiple OLS regression

Model 1: $yearsgap_i = \varphi_0 + \varphi_1 female_i + u_i$

Model 2: $yearsgap_i = \alpha_0 + \alpha_1 female_i + \alpha_2 age_i + v_i$

Estimation coefficient results

The coefficient of females in simple OLS regression suggests that females have educational gaps 0.9890 years larger than males on average. The estimated magnitude is economically and statistically very significant; each year of schooling predicts an increase in lifetime earnings by 10.5% in sub-saharan Africa (Psacharopoulos & Patrinos, 2018), which has large implications in building generational wealth. This relationship persists when controlling for age.

Methodology - Mediator: IV and 2SLS

Model 3:

$$years_{gap_i} = \beta_0 + \beta_1 agrit_i + \beta_2 female_i + \beta_3 age_i + V_i$$

Our choice mediator channel for investigation is agricultural participation ($agrit_i$), which also proxies for wealth in rural areas. I aim to extract the specific gender effect through agricultural participation, using the difference of $female_i$ coefficients in models 3 and 4. α_1 captures all gender effects, whereas β_2 captures only non- $agrit$ gender effects, and thus the difference is gender's effect through said mediator.

I use a subsample of 10-25 year old individuals who are children of the household heads, avoiding younger children who are not in school or work due to age, and avoiding individuals past school age to measure gender's effect on current educational decisions.

IV and 2SLS

Our mediator is agricultural labour participation, and I experiment with $wallhat_i$ and $foodinsec_i$ as our instrumental variables and run IV and 2SLS using each variable for $agrit_i$.

Foodinsec IV (Dummy for food insecurity experienced in past 12 months)

$$\text{Stage 1: } \widehat{agrit_i} = \theta_0 + \theta_1 foodinsec_i + \theta_2 female_i + \theta_3 age_i + U_i$$

$$\text{Stage 2: } years_{gap_i} = \beta_0 + \beta_1 \widehat{agrit_i} + \beta_2 female_i + \beta_3 age_i + V_i$$

Wallhat IV (Dummy for wall material, permanent vs non-permanent materials)

$$\text{Stage 1: } \widehat{agrit_i} = \theta_0 + \theta_1 wallhat_i + \theta_2 female_i + \theta_3 age_i + W_i$$

$$\text{Stage 2: } years_{gap_i} = \beta_0 + \beta_1 \widehat{agrit_i} + \beta_2 female_i + \beta_3 age_i + Q_i$$

Exogeneity and Relevance

Relevance for $wallhat_i$ is empirically evidenced through first stage F-statistics that pass the Stock-Yogo threshold of 10, and theoretically, households with permanent walls are representative of wealth and land ownership in rural agricultural areas, which is correlated with agricultural participation.

Relevance for $Foodinsec_i$ is empirically weaker with a 6.69 first stage F-statistic, and is negatively associated with $agrit_i$, as poorer households are more likely to be food insecure and less likely to participate in agricultural labour.

Exogeneity for these two variables is similarly argued, whilst there is plausible correlation between the instrumental variables and unspecified channels affecting $yearsgap_i$ such as wealth, hunger and location, these effects are argued to affect individuals equally regardless of gender, and only bias the $agrit_i$ estimator rather than the gender specific effect.

$wallhat_i$'s Wu-Hausman values being strong across all subsamples implies sufficient variation separate from any endogenous components, and the stronger instrument ($F = 181.9^{***}$) in $wallhat_i$ compared to $foodinsec_i$ resulting in a stronger amplification of magnitude of $agrit_i$'s coefficient reflects meaningful distinction between instruments and the unspecified variables in the error term.

Results - IV/2SLS

Estimation Coefficient Results - Child of Household Head Age 10-25

Variable	Reduced OLS	Stage 1 - Foodinsec	Stage 2 - Foodinsec	Stage 1 - Wallhat	Stage 2 - Wallhat	OLS with Agrit
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Constant	-2.6330*** (0.43957)	0.6624*** (0.029792)	-2.71311*** (0.44261)	0.50242*** (0.53357)	-3.02009*** (0.53357)	-2.7444*** (0.4193)
Female	-0.86788** * ()	0.0735*** (0.014698)	-0.67268. (0.37363)	0.05533*** (0.22905)	0.07533 (0.22905)	-0.5964*** (0.1688)
Age	0.48048*** (0.02197)	-0.0090*** (0.001715)	0.50903*** (0.05315)	-0.01119*** (0.02940)	0.61842*** (0.02940)	0.5202*** (0.0212)
Agrit			-1.46422** * (2.50345)		-7.07486*** (0.64203)	-2.0367*** (0.1668)
IV (foodinsec _i or wallhat _i)		-0.0826*** (0.016194)		0.32476*** (0.01387)		
N	1492	4674	1492	4674	1492	1492
R ²	0.2528	0.01634	0.315		-0.09615	0.3204
Weak instruments			6.690**		181.9***	
F statistic	253.2	26.87		203.1		235.3
Wu-Hausman			0.81810		120.4***	

Adding agricultural labour using foodinsec_i via IV and 2SLS, I see women have an educational gap 0.673 years smaller than mens', losing statistical significance at the 5% level ($p = 0.072$). I would take $(\alpha_1 - \beta_2) = (-0.1952)$ to be the effect of the mediator, but β_2 loses significance in the 2SLS model due to the weak instrument (F-statistic diagnostic from ivreg = 6.690).

Sons in this 10-25 age sample, are experiencing an average educational gap of 0.9228 years, of which 0.1952 would have been explained by disproportionate participation in agricultural labour, but these IV estimates should be interpreted cautiously.

The dependent variable yearsgap is a cumulation of many individual and time specific effects over one's life, and gender's mediating effects would be similarly cumulative. Thus extracting exogeneity and a mediating effect would require panel data over said periods. Searching for a

suitable IV requires it can extract a gender specific effect whilst fulfilling the exclusion criteria of not resulting in gender specific bias, this conflicting trade off is what makes finding a suitable IV in cross-sectional data extremely hard, and why panel data is necessary to quantify mediator channels for gender on educational gaps.

I attempted to circumvent this limitation by using $wallhat_i$ as an IV, given that housing material is more similar to $years_{gap}$ in its specification than $foodinsec$, but this IV also falls short in extracting a specific gender mediator effect in this sample, and returns a much less significant ($p = 0.742$) $female_i$ coefficient. Thus I interpret $(\alpha_1 - \beta_2) = (-0.1952)$ ($p = 0.072$) as an imprecise estimator given its limitations with the temporal misspecification of the data.

In the first stage of 2SLS with $foodinsec_i$, I see agricultural participation is increasing for females (0.073510^{***}). The F-statistic returns well above 10, indicating a strong instrument, but the weak instruments test result of 6.69 is under the Stock-Yogo threshold of 10, which causes it to fall short in quantifying a significant mediator effect.

The total gender effect on $years_{gap}$, however remains causally identified in this cross sectional data due to random assignment of gender.

Extensions and Robustness checks

Extension 1 - Split OLS at age 25

As the pooled OLS estimation may mask important heterogeneity across age groups, I split the sample at age 25 to examine whether the gender effect differs across school-age vs older individuals. Since the benchmark schooling measure is mechanically capped after age 25,

individuals who are still accumulating yearsgap_i may experience different effects to those who have finished accumulating.

Model 4:

$$\text{yearsgap}_i = \varphi_0 + \varphi_1 \text{female}_i + u_i$$

Estimation Coefficient Results

Variable	Below 25	Above 25
Female	-0.3129** (0.10836)	1.1602*** (0.083460)
Constant	-3.4982*** (0.23412)	6.8464*** (0.139404)
Age	0.5311*** (0.01159)	0.0981*** (0.002882)
N	3589	8096
R ²	0.369	0.1454

For individuals aged 25 or below, the female coefficient is -0.313 ($p < 0.001$), younger females have a 0.313 years smaller education gap than males within the current school-age generation. For individuals above age 25, the coefficient becomes +1.16 ($p < 0.0001$), indicating that older females have a larger education gap of 1.16 years relative to older males.

Robustness Check - Alternative Age Threshold

To check whether the sign reversal depends on the age cut-off or not, I conduct a robustness check splitting at age 22 instead of 25.

Model 5:

$$yearsgap_i = \Phi_0 + \Phi_1 female_i + k_i$$

Estimation Coefficient Results

Variable	Below 22	Above 22
Female	-0.29333** (0.12262)	0.9857*** (0.0783)
Constant	-2.42502*** (0.26278)	6.78264*** (0.12188)
Age	0.45888*** (0.01442)	0.10107*** (0.002607)
N	2550	9135
R ²	0.284	0.1555

In this subsample, estimators are similar to the split subsample at 25 years and maintain statistical significance, suggesting that the heterogeneity pattern is not driven by the specific choice of age cut-off, but demonstrates a more robust difference across age cohorts.

I attribute this subsample effect to the younger subsample estimating the effect of gender on ongoing and recent schooling decisions, whereas the older subsample represents the total effect of gender bias on individuals who have already finished accumulating their educational gap. This effect is consistent with young women being more likely to stay in school, but older women were systematically disadvantaged in terms of educational access, and the pooled OLS acts as a weighted average of these two effects.

Extension 2 - IV and 2SLS - aged 25-45 household heads, Model 3.

I instrument Agrit using wallhat_i; again, using an older subsample, 25-45 year old household heads, to see if agricultural participation affects non-school age individuals differently, in an

environment where agricultural labour is likely to serve as a primary income source rather than a complement to ongoing education.

Variable	Reduced OLS	Stage 1 - Wallhat	Stage 2 - Wallhat	Reduced OLS With Agrit
Constant	6.82221*** (0.48541)	0.460253*** (0.058440)	12.29064*** (0.62253)	8.83581*** (0.47536)
Female	0.97512** (0.20681)	0.109703*** (0.024105)	1.59865*** (0.22721)	
Age	0.10347*** (0.01365)	-0.007867*** (0.001587)	0.04063** (0.01527)	0.08777*** (0.01292)
Agrit			-6.84775*** (0.41737)	-2.59692*** (0.15270)
IV (wallhat _i)		0.438906*** (0.019725)		
N	2307	2342	2307	2307
R ²	0.03592	0.1862	-0.1312	0.1352
Weak instruments			491.8***	
F statistic	43.96	179.6		181.2
Wu-Hausman			174.3***	

Results

I see that in this 25-45 sample, the IV is much stronger, passing the F-statistic test (>10), the weak instruments and Wu-Hausman test. I see that the female coefficient is now significant at the 0.1% level.

Out of school individuals are no longer accumulating yearsgap and wall material is a slower changing IV that better maps to the educational choices that the household head faced in their formative years.

Thus I am able to use the $\alpha_1 - \beta_2$ formula to extract the mediator's effect with significantly more confidence than before: (-0.624) . Women in this subsample on average experience a .975 year disadvantage in educational gap, females having a higher agricultural participation rate (female_i in stage 1 = 0.1097***) offsets this by 0.624 years, helping to bridge the educational gap.

Agricultural participation proxies for rural land and asset ownership, which is positively correlated with educational outcomes, as shown in the OLS (agrit's coefficient = -2.59692***). This interpretation however, is still not perfectly specified for causal interpretation due to agrit_i's issues in specification and whilst wall material is a slow changing variable, it may have been different when the household head was of school-age vs time of survey.

Conclusions

In conclusion, our main OLS estimation shows that gender has considerable effect on educational gaps, even after controlling for age, females on average have a larger educational gap than men. These effects are not uniform by age as pool OLS hides cohort differences: young females have smaller gaps than young males, while in older generations, females have a larger shortfall relative to males. The sign reversal remains when changing the age cut-off from 25 to 22, so our findings do not merely depend on the specific chosen age cut-off.

Because gender is a predetermined characteristic, I examine the effect of it through a mediator channel - agricultural participation. In order to identify this mediator, food insecurity and wall material are used as IVs. While I am able to quantify this mediator channel for an older subsample, these should be interpreted cautiously due to cross-sectional data limitations and instrument validity. In the household sample aged 25-45, using wall material as an instrumental variable shows this channel helps lessen female household heads' educational disadvantage.

Cross-sectional data does not track individuals over time, making it difficult to isolate causal mediator effects exactly when education decisions were made. Moreover, policy should not only consider the overall gender gap, but must be tailored to the demographic they aim to target. For further research on educational outcomes, panel data must be gathered for mediator distillation.

References

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APPENDIX: R CODE

```
library(AER)
setwd("C:/Users/xqh02/Desktop/ECONOMETRICS/ASSIGNMENT 1")
a1data2 <- read.csv("a1data2.csv")
a1data2above7 <- subset(a1data2, age >= 7)

#Summary Statistics

mean(a1data2above7$yearsedu, na.rm = TRUE)
sd(a1data2above7$yearsedu, na.rm = TRUE)
range(a1data2above7$yearsedu, na.rm = TRUE)

median(a1data2above7$yearsgap, na.rm = TRUE)
```

```
mean(a1data2above7$yearsgap, na.rm = TRUE)
sd(a1data2above7$yearsgap, na.rm = TRUE)
range(a1data2above7$yearsgap, na.rm = TRUE)
```

```
mean(a1data2above7$female, na.rm = TRUE)
sd(a1data2above7$female, na.rm = TRUE)
range(a1data2above7$female, na.rm = TRUE)
```

```
mean(a1data2above7$age, na.rm = TRUE)
sd(a1data2above7$age, na.rm = TRUE)
range(a1data2above7$age, na.rm = TRUE)
```

#Simple OLS

```
eq1 <- lm (yearsgap~female+age, data=a1data2above7)
print(summary(eq1))
nobs(eq1)
eq2 <- lm (yearsgap~age, data=a1data2above7)
print(summary(eq2))
eq3 <- lm (yearsgap~female, data=a1data2above7)
print(summary(eq3))
nobs(eq3)
```

#split subsamples @25

```
a1data2above7_below25 <- subset(a1data2above7, age <= 25)
a1data2above7_above25 <- subset(a1data2above7, age > 25)
eq_below25 <- lm(yearsgap ~ female + age, data = a1data2above7_below25)
eq_above25 <- lm(yearsgap ~ female + age, data = a1data2above7_above25)
print(summary(eq_below25))
print(summary(eq_above25))
nobs(eq_below25)
nobs(eq_above25)
```

#split subsamples @22 (robustness)

```
a1data2above7_below22 <- subset(a1data2above7, age <= 22)
a1data2above7_above22 <- subset(a1data2above7, age > 22)
eq_below22 <- lm(yearsgap ~ female + age, data = a1data2above7_below22)
eq_above22 <- lm(yearsgap ~ female + age, data = a1data2above7_above22)
print(summary(eq_below22))
print(summary(eq_above22))
```



```
nobs(eq_below22)
nobs(eq_above22)
```

```
#mapping foodinsec to all individuals within a household
hh <- read.csv("hh_sec_h.csv")
hh$foodinsec <- ifelse(hh$hh_h08 == 1, 1, 0)
foodinsec_hh <- hh[, c("y5_hhid", "foodinsec")]
a1data2above7 <- merge(a1data2above7, foodinsec_hh, by = "y5_hhid", all.x = TRUE)
```

```
#mapping wall material to all individuals within a household
ww <- read.csv("hh_sec_i.csv")
ww$wallhat <- ifelse(ww$hh_i08 %in% c(5, 6), 1, 0)
wall_hh <- ww[, c("y5_hhid", "wallhat")]
a1data2above7 <- merge(a1data2above7, wall_hh, by = "y5_hhid", all.x = TRUE)
```

```
#son or daughter of head between 15-25
a1data2above7_child1025 <- subset(a1data2above7, hh_b05 == 3 & age >= 10 & age <= 25)
a1data2above7_child1022 <- subset(a1data2above7, hh_b05 == 3 & age >= 10 & age <= 22)
a1data2above7_head2545 <- subset(a1data2above7, hh_b05 == 1 & age >= 25 & age <= 45)
```

```
#testing mediator = agrit (1 if yes, 0 if no) - child1025
eq_stage1f <- lm(agrit ~ foodinsec + female + age,
  data = a1data2above7_child1025)
print(summary(eq_stage1f))
nobs(eq_stage1f)
```

```
eq_3sls <- ivreg(yearsgap ~ agrit + female + age |
  foodinsec + female + age,
  data = a1data2above7_child1025)
print(summary(eq_3sls, diagnostics = TRUE))
nobs(eq_3sls)
```

```
# OLS for same sample
eq_reduced <- lm(yearsgap ~ female + age, data = a1data2above7_child1025)
print(summary(eq_reduced))
nobs(eq_reduced)
```

```
eq_reduced2 <- lm(yearsgap ~ agrit + age, data = a1data2above7_child1025)
print(summary(eq_reduced2))
```

```

# OLS on agrit
eq_agrit_ols <- lm(yearsgap ~ agrit + female + age, data = a1data2above7_child1025)
print(summary(eq_agrit_ols))
nobs(eq_agrit_ols)

#testing mediator = wallhat (1 if permananet, 0 if impermanent) - child 1025
eq_stage1w <- lm(agrit ~ wallhat + female + age,
  data = a1data2above7_child1025)
print(summary(eq_stage1w))
nobs(eq_stage1w)

eq_3slsw <- ivreg(yearsgap ~ agrit + female + age |
  wallhat + female + age,
  data = a1data2above7_child1025)
print(summary(eq_3slsw, diagnostics = TRUE))
nobs(eq_3slsw)

#wallhat robustness check - head2545
eq_hwwstage1f5 <- lm(agrit ~ wallhat + female + age,
  data = a1data2above7_head2545)
print(summary(eq_hwwstage1f5))
nobs(eq_hwwstage1f5)

eq_hww3sls5 <- ivreg(yearsgap ~ agrit + female + age |
  wallhat + female + age,
  data = a1data2above7_head2545)
print(summary(eq_hww3sls5, diagnostics = TRUE))
nobs(eq_hww3sls5)

eq_reduced_head2545 <- lm(yearsgap ~ female + age,
  data = a1data2above7_head2545)
print(summary(eq_reduced_head2545))
nobs(eq_reduced_head2545)

eq_reduced_head2545agrit <- lm(yearsgap ~ agrit + age, data = a1data2above7_head2545)
print(summary(eq_reduced_head2545agrit))

```